
Machine Learning with applications to multi-omics and clinical prediction

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This course

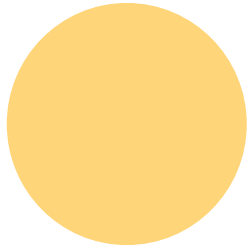
- **Will not teach you specific research pipelines or how to use particular tools**
- **Technical descriptions of some broad ideas to give general insights**
- **Mathematically demanding (linear algebra, numerical optimisation, probability)**
- **We assume familiarity with Python**

This course

- **Mornings: Lecture material + Group Exercises**
- **Afternoon: Self-paced practicals (in Python)**

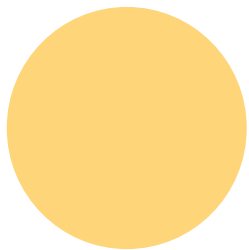
Schedule

Day	Session	Topic	Format
Wednesday	Morning	Supervised Learning	Lectures
Wednesday	Afternoon	Supervised Learning	Practical
Thursday	Morning	Unsupervised Learning	Lectures
Thursday	Afternoon	Unsupervised Learning	Practical
Friday	Morning	Self-Supervised Learning / Research Talks	Lectures



What is multi-omics?





What is multi-omics?

In modern usage, it is a generic term to describe a range of high-dimensional measurement modalities.



Multi-omics

Traditional

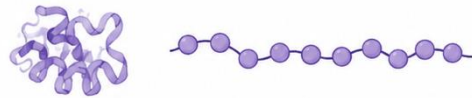
Genomics – DNA



Transcriptomics - RNA



Proteomics – Proteins



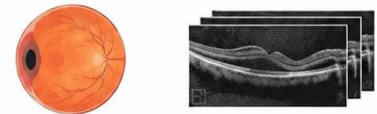
Metabolomics - Metabolites



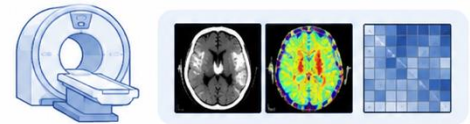
Molecular measurements

Recent examples

Oculomics – eye scans



Radiomics – radioimaging

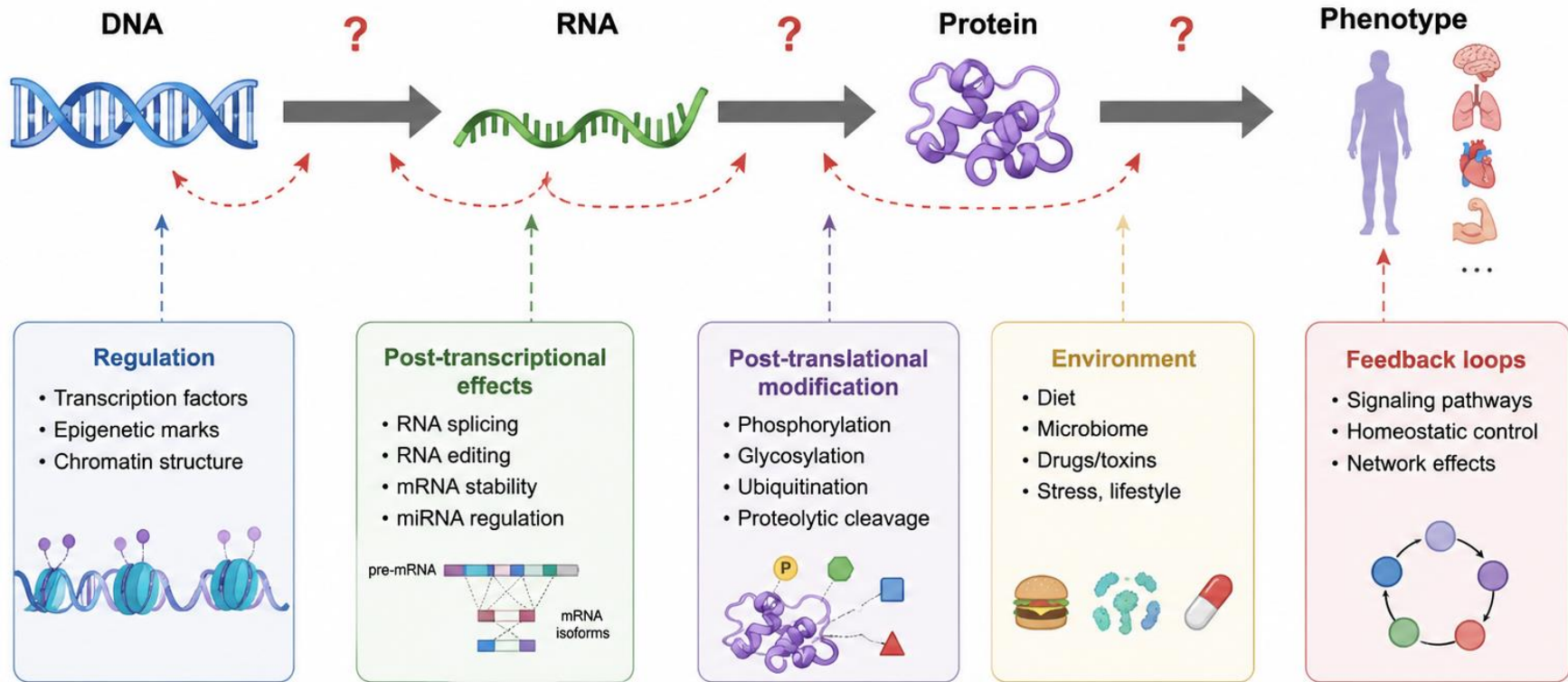


Phenomics – deep phenotyping



High-dimensional measurements

Why do we need multiple omics?

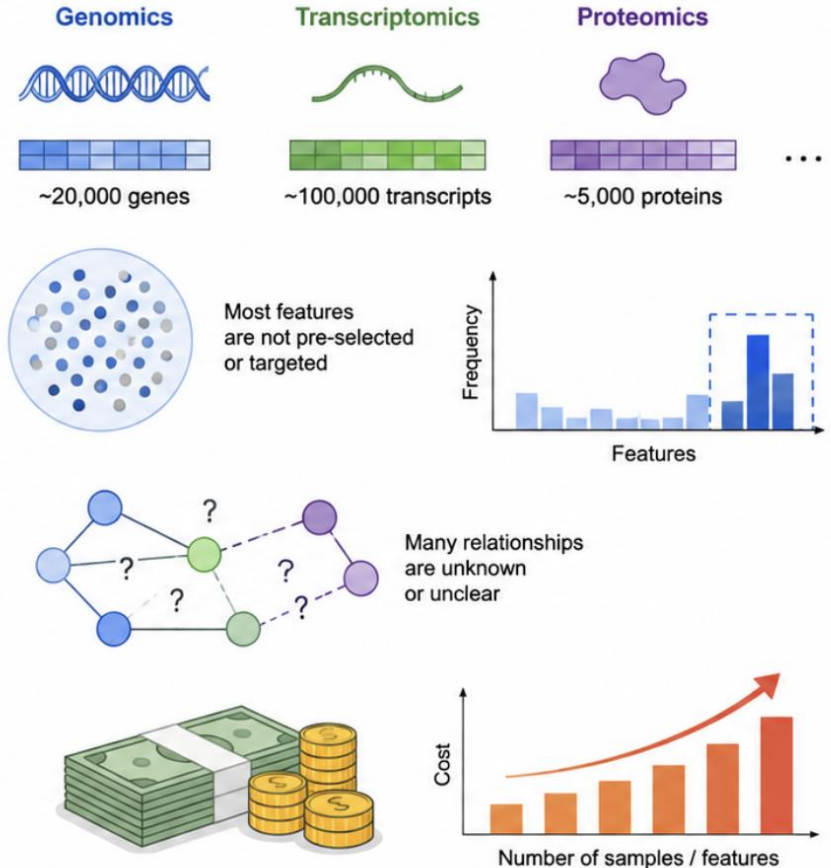


If transcriptomics perfectly predicted proteins and proteins perfectly predicted phenotype, we would not need multi-omics.

Properties of multi-omics

Each 'omics has:

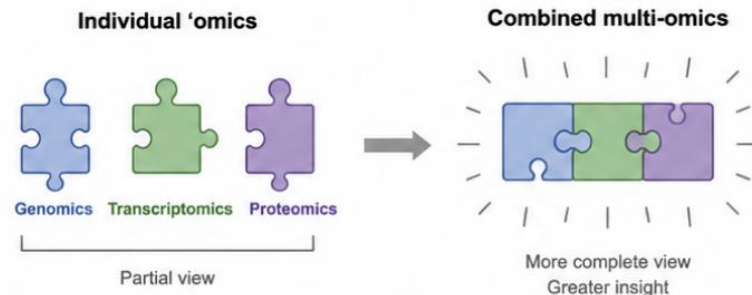
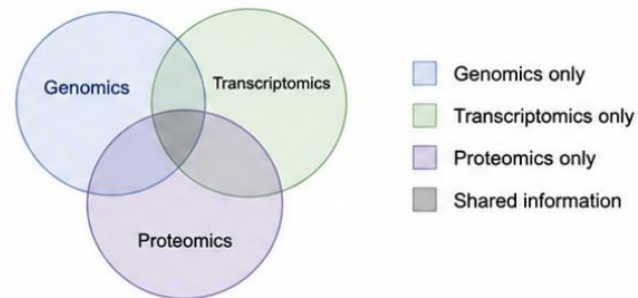
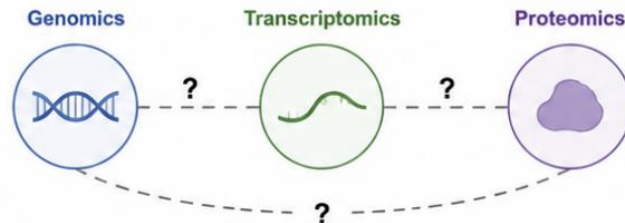
- Large numbers of features
- Untargeted features
- Poorly understood relationship between features
- Deep information about a particular aspect of biology or behaviour
- Expensive!



Properties of multi-omics

Across 'omics:

- Relationships between 'omics are not understood
- Information content is not orthogonal
- Hypothesis that combination is greater than sum of individual 'omics

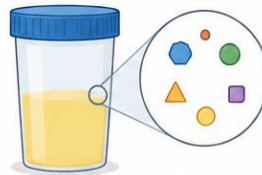


Examples

Orthogonal: Tissue-specific data

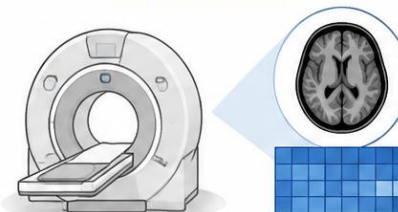
- Metabolites from urine
- Imaging of brain tissue

Urine metabolomics



Metabolites from urine

Brain radiomics



Imaging of brain tissue

Overlap: Data from a potentially causal pathway

- Blood-based protein levels
- Liver-based transcriptomics

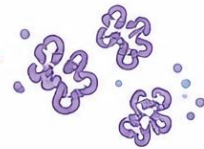
Liver → Proteins → Blood
(potentially causal pathway)

Liver transcriptomics



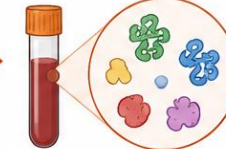
Liver-based transcriptomics

Protein production

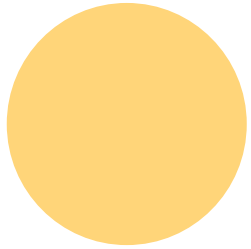


Proteins synthesized in the liver

Blood proteomics



Blood-based protein levels



Understanding multi-omics



Understanding multi-omics

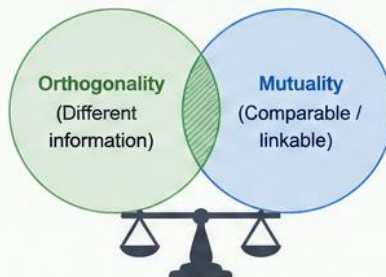
1 Require a great deal of thought



- ✓ Start with a clear biological question
- ✓ Define hypotheses and expected relationships
- ✓ Plan the whole study end-to-end

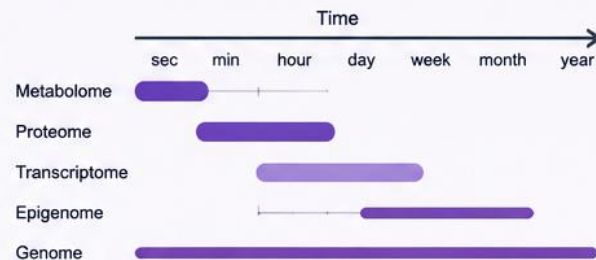
Design drives insight.

2 Balance orthogonality and mutuality



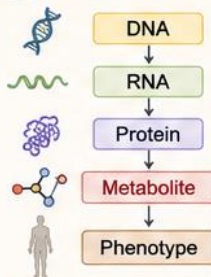
Maximize complementary information while ensuring integration is possible.

3 Time resolution



Match sampling frequency to the dynamics of each layer.

4 Mechanistic plausibility

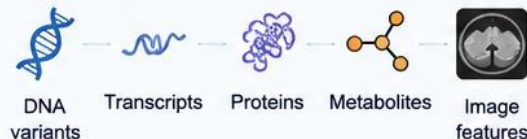


Do the proposed links make biological sense?

- Known pathways?
- Regulatory logic?
- Directionality?
- Timescale consistency?

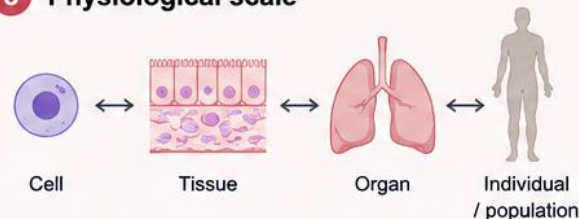
Build designs grounded in biology.

5 Measurement units



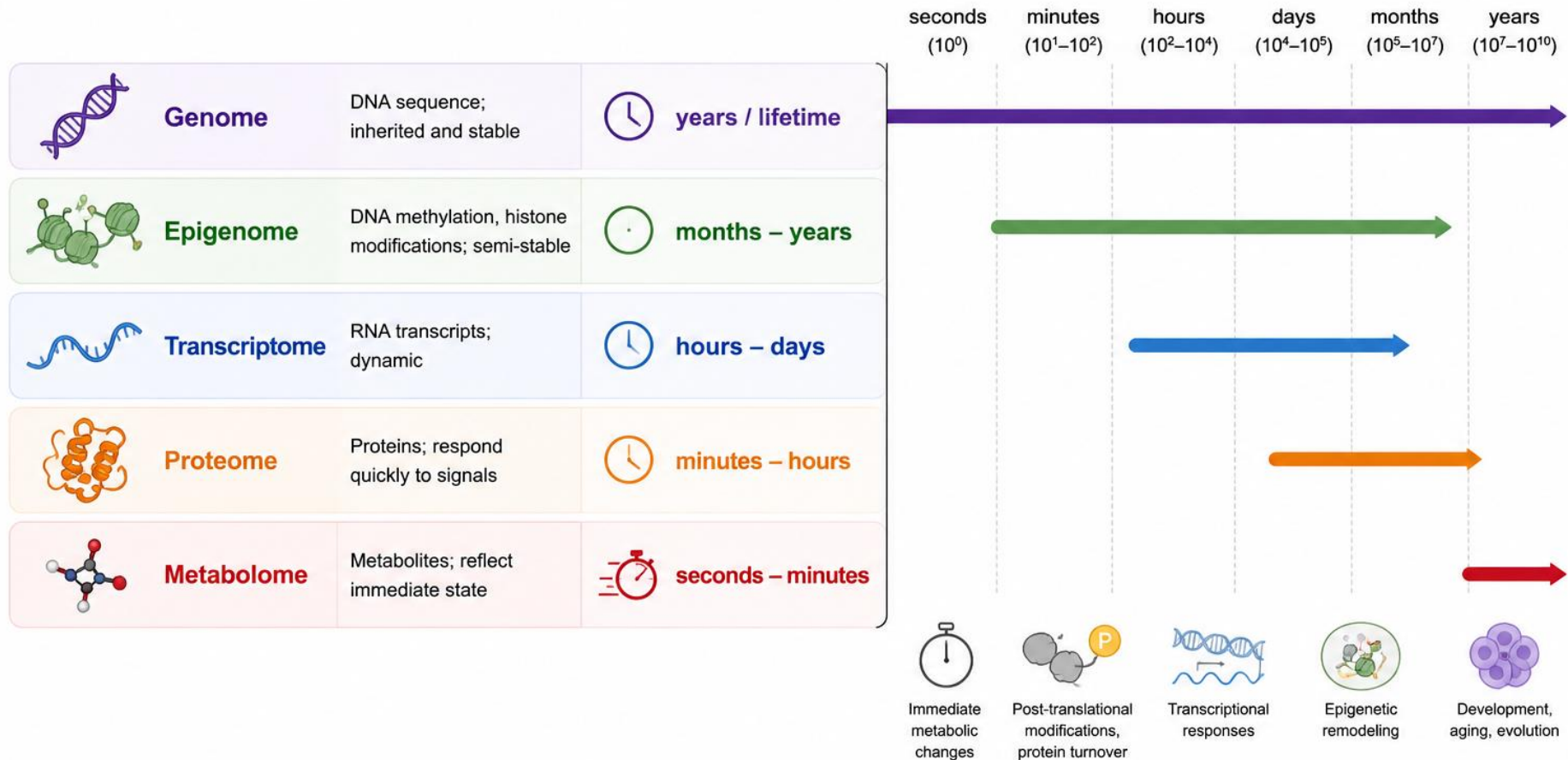
Different data types = different units.
Plan for how they will be measured, normalized and integrated.

6 Physiological scale



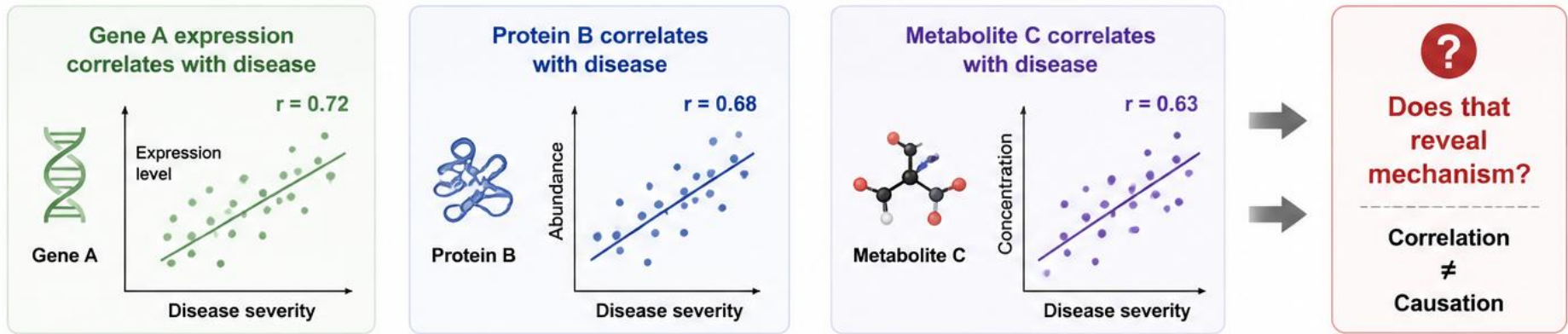
Ensure the chosen scale(s) align with the biology of interest and the study question.

Time scales



Mechanistic interpretation

Statistical analysis can reveal relationships between measurements and outcome measures.

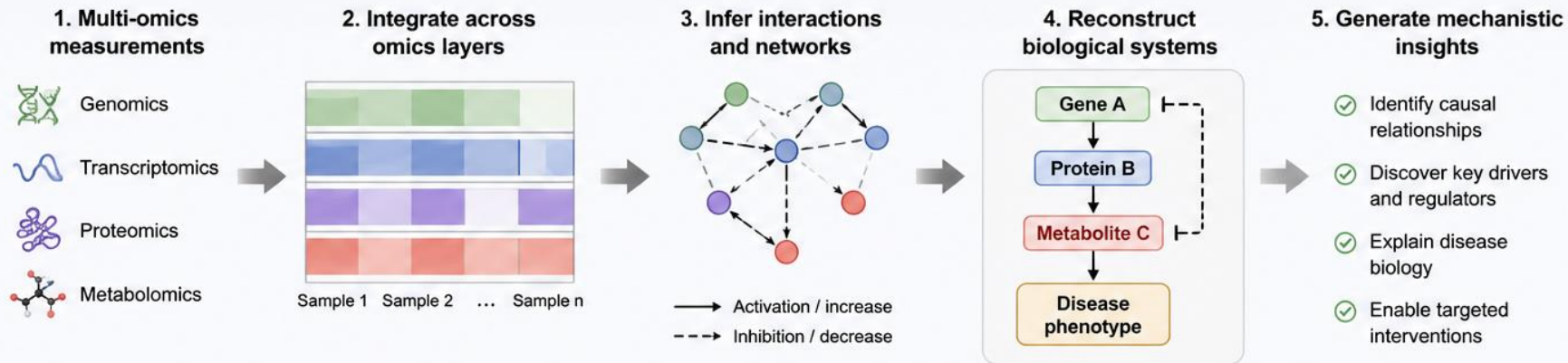


Without mechanistic interpretation, multi-omics only finds us correlative biomarkers.

Mechanistic interpretation

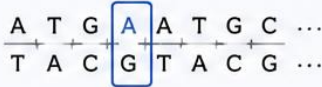
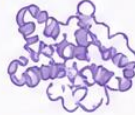
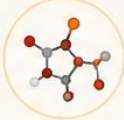
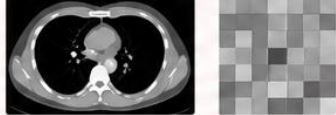
Multi-omics information should ideally be used to inform mechanistic models.

From correlation to mechanism: reconstructing biological systems

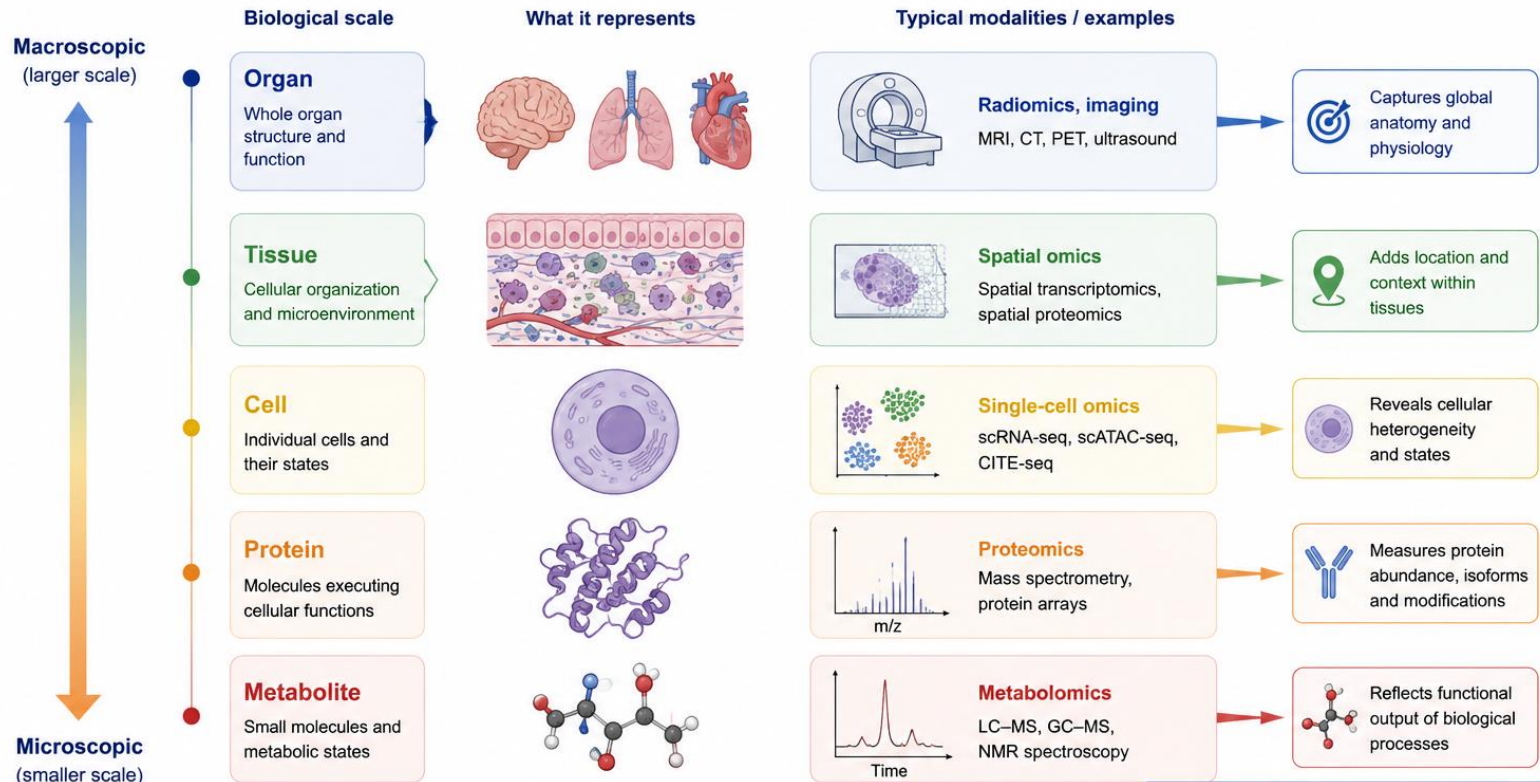


In practice, we have not consistently developed this type of general capability.

Unit of measurement

Modality	Unit	What it represents
 Genomics	 DNA variants	Differences in the DNA sequence between individuals
 Transcriptomics	 transcripts	RNA molecules produced by genes
 Proteomics	 proteins	Proteins expressed and functioning in the system
 Metabolomics	 metabolites	Small molecules and metabolic intermediates
 Radiomics	 image features	Quantitative features extracted from medical images

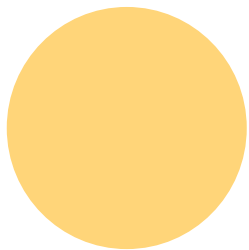
Scale of measurement



Are measurements made at different scales answering the same biological question?



Multi-omics integration requires connecting measurements across multiple scales of biological organization.
Each scale provides unique and complementary information—but also different challenges.

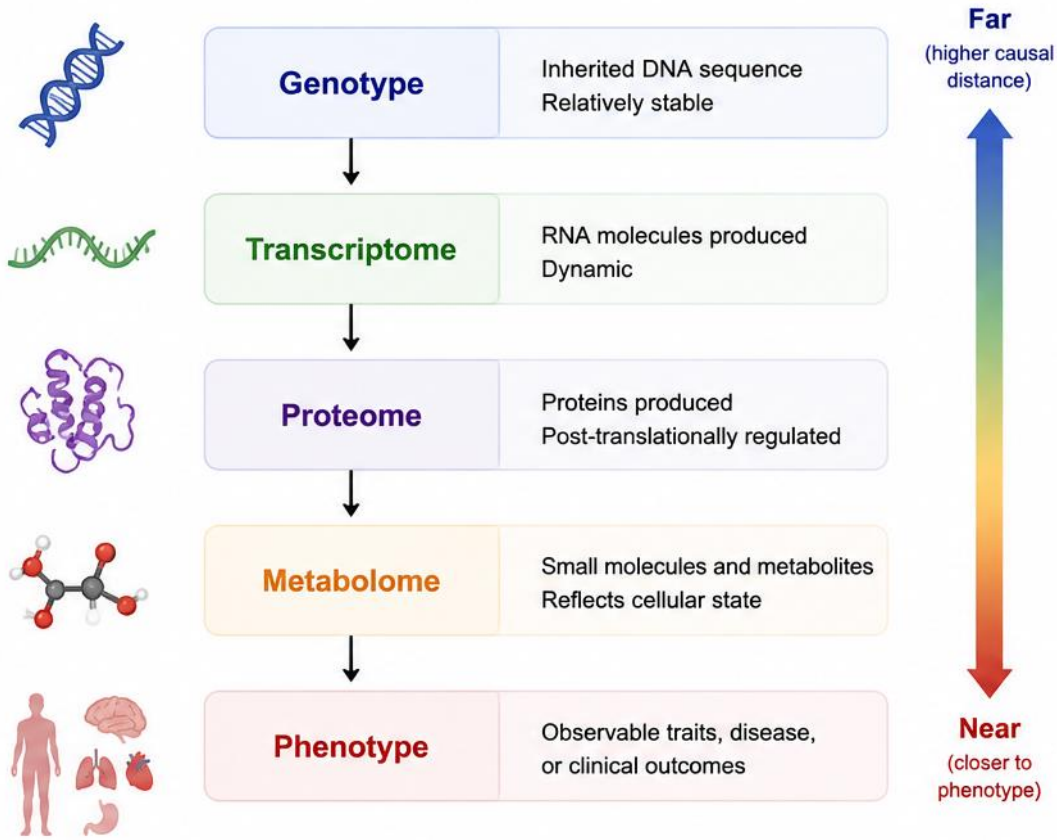


Causal distance

Something we do not talk about enough

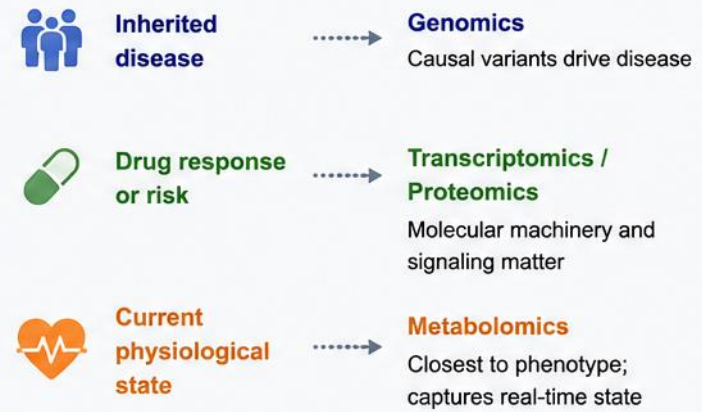


Causal distance



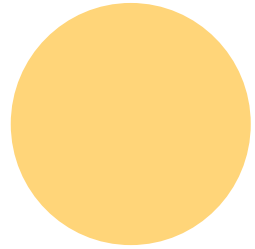
Which layer is most predictive of disease?

It depends.



The closer a measurement is to the phenotype, the more factors can obscure the signal.

Upstream measurements may be more stable but less directly predictive.



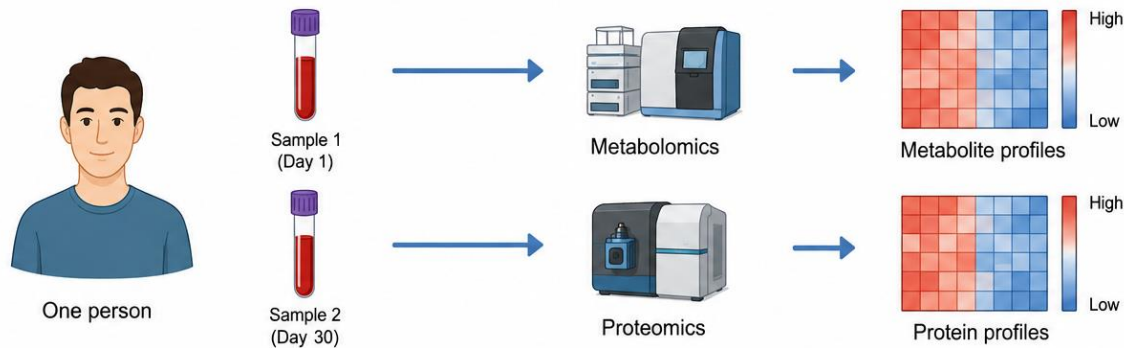
Multi-omics experiment design



Experiment design examples

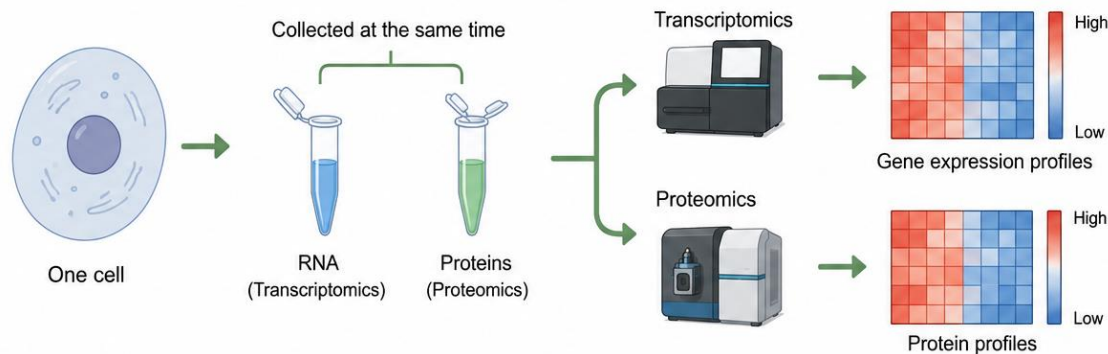
1. Multiple independent samples from the same common object

- One person
- Metabolomics, Proteomics
- Different blood samples



2. Multiple paired samples from the same object

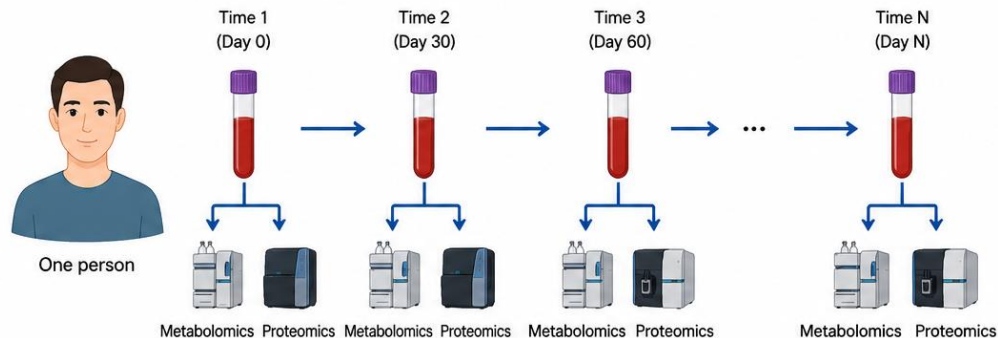
- One cell
- Transcriptomics, Proteomics
- Collected at the same time



Experiment design examples

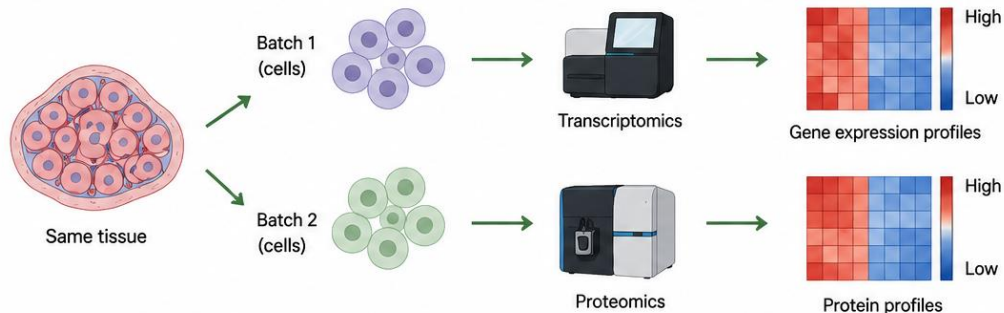
3. Longitudinal sampling from the same common object

- One person
- Different blood samples collected over time
- Metabolomics, Proteomics generated at each time point



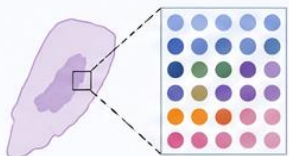
4. Modality-specific sampling from different objects

- Two batches of cells drawn from the same tissue
- One batch is measured transcriptomics, the other for proteomics



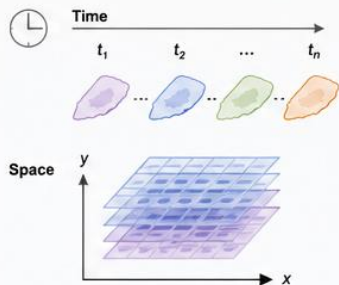
The spatial dimension

1 Emergence of spatial omics technologies



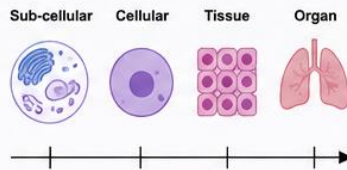
Recent years have seen the emergence of spatial omics technologies

2 Measurements in both time and space



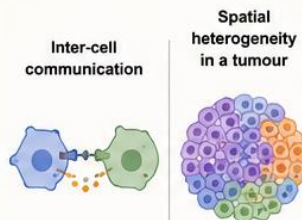
Ability to locate measurements in both time and space

3 Scale matters



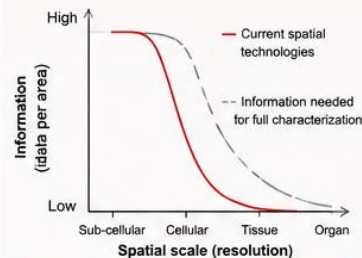
Does the spatial scale matter for the problem being investigated?

4 Different questions require different spatial scales



For example, inter-cell communication vs spatial heterogeneity in a tumour

5 Current technologies – information depth depends on scale

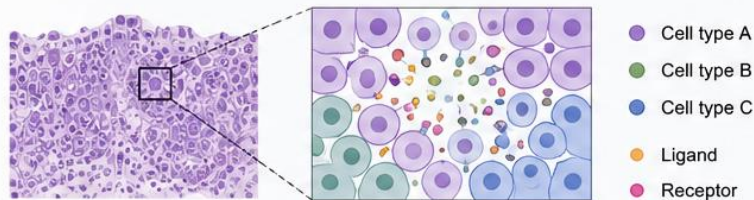


Currently spatial technologies only generate enough information for the former, but not the latter

The spatial dimension

Example 1: Inter-cell communication

Needs single-cell (or near single-cell) resolution

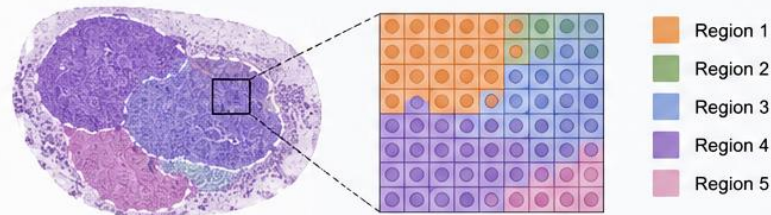


High resolution is needed to detect cell–cell interactions and communication networks.

vs.

Example 2: Spatial heterogeneity in a tumour

Needs larger field of view (lower resolution)

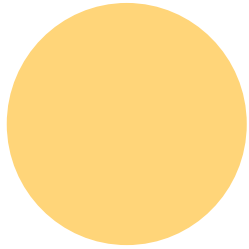


Lower resolution over a large area is needed to capture overall spatial patterns and heterogeneity.



Spatial omics adds the dimension of “where” to molecular measurements.

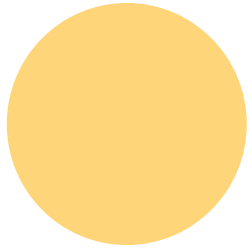
The right spatial scale depends on the biological question—and current technologies are better suited for inter-cell communication than for large-scale spatial heterogeneity.



Multi-omics integration

What does it mean to “integrate” multi-omics data?



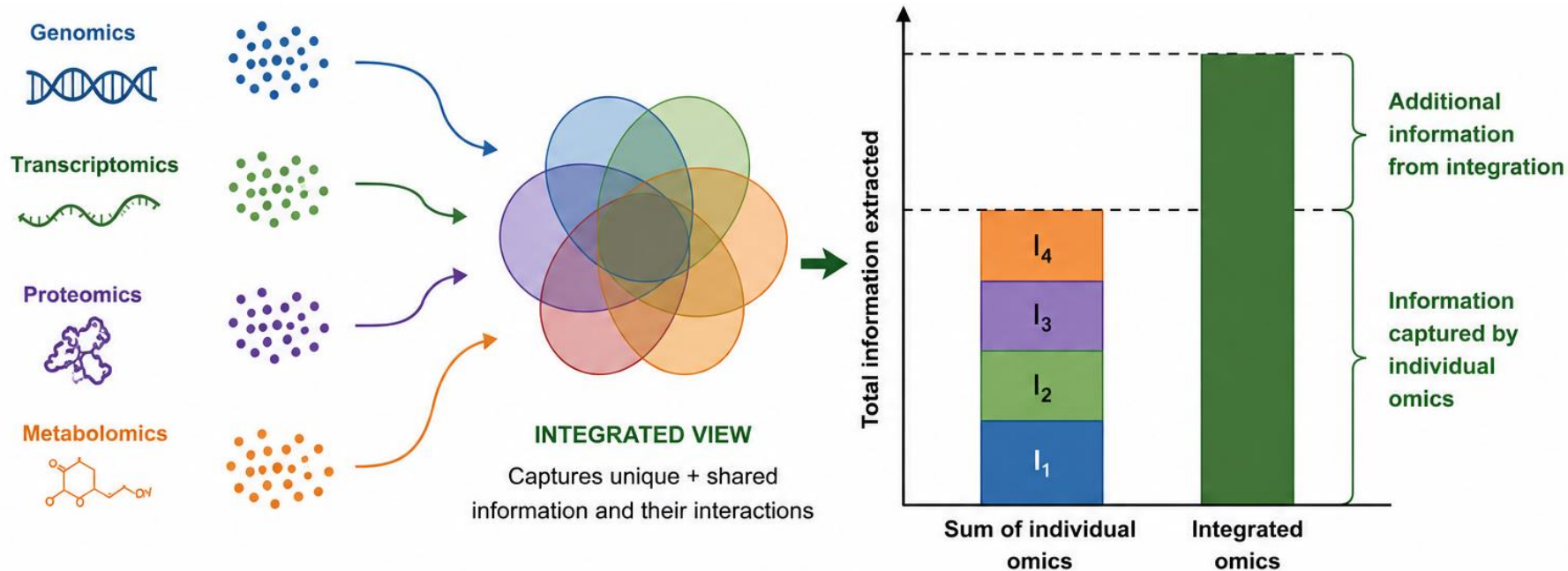


Multi-omics integration

Harness information content from across different modalities to gain insight into the overall system

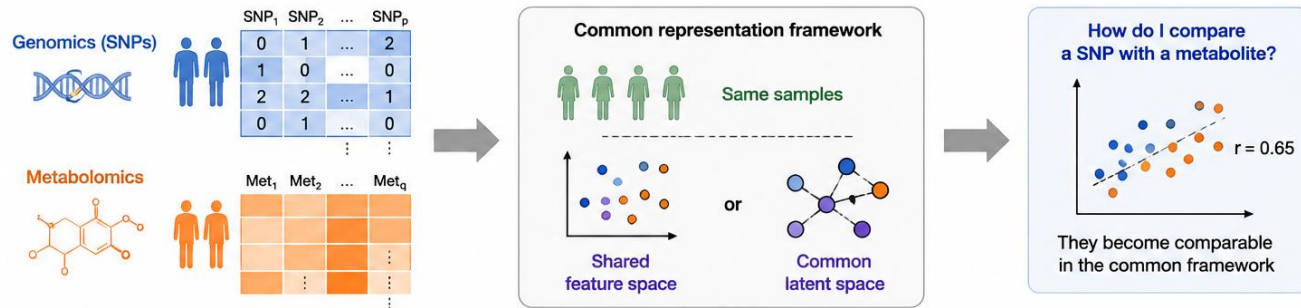


What does it mean to integrate?

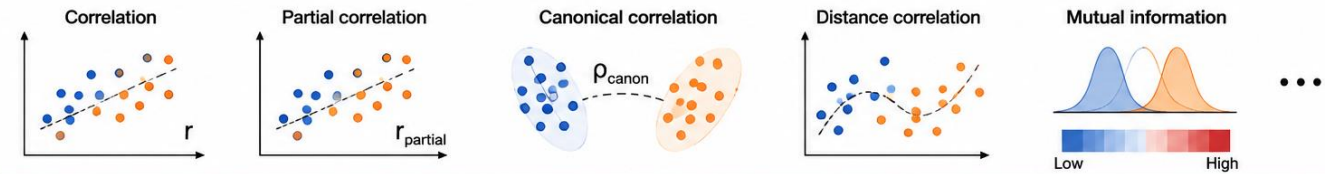


How to integrate

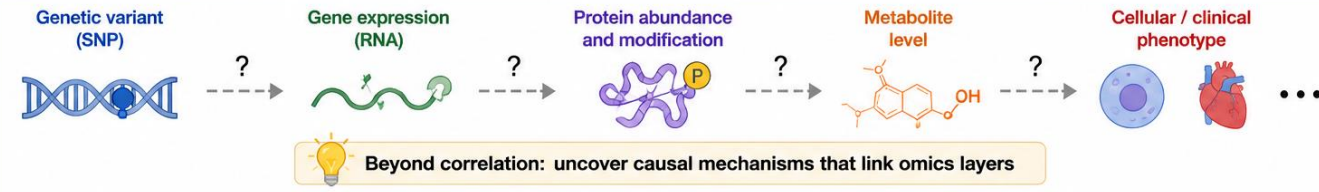
1. Alignment: put different data modalities into a common framework



2. Typically via statistical correlation-linked measures



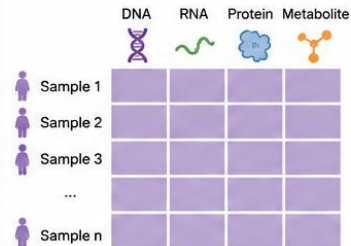
3. Under-explored area: mechanistic linkage



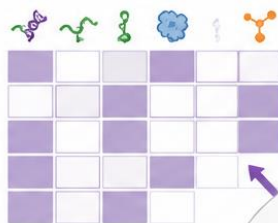
What makes it challenging

1. Generally imperfect experimental designs

Ideally: all samples measured for all omics



Reality: missing modalities and samples



2. Compromises

More samples



Single omic



Lower cost

More omics



Fewer samples



Higher cost

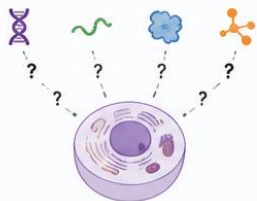
VS.



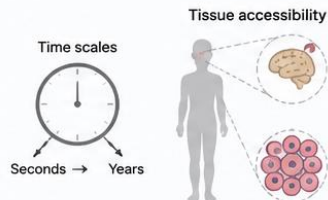
Limited resources force trade-offs between depth (omics) and breadth (samples)

3. Biological constraints

Inability to measure more than one thing at the same time on the same object



Accessibility of tissue or time scales of molecular activity

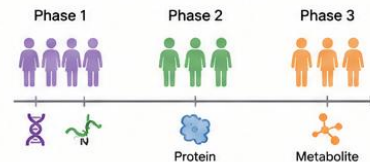


4. Technical constraints

Cost



Phased designs leading to incomplete observations



Different samples at different phases
→ incomplete multi-omics view

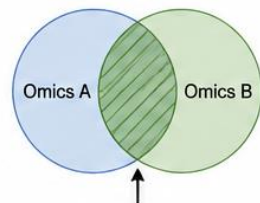
Why multi-omics integration is challenging



Does more always mean better?

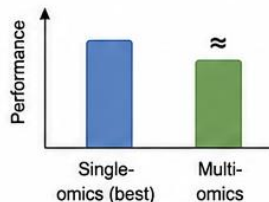
If more information is better, does multi-omics always outperform single-omics?

1 Redundant information



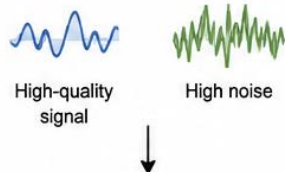
Large overlap → little new information gained

Predictive performance



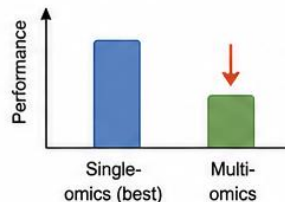
Little or no improvement

2 Noisy modalities



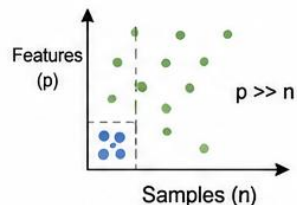
Noise adds uncertainty and can obscure true signal

Predictive performance



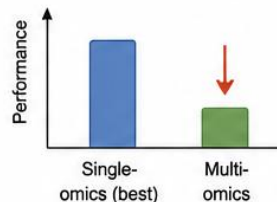
Performance can decrease

3 Small sample sizes



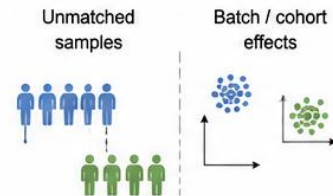
High dimensionality with few samples → overfitting risk

Predictive performance



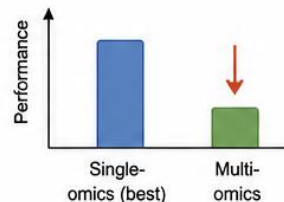
Overfitting → worse performance

4 Poor study design



Confounding and batch effects mask true biological signal

Predictive performance



Spurious signals dominate

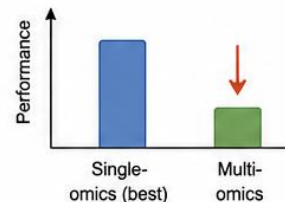
5 Wrong biological question



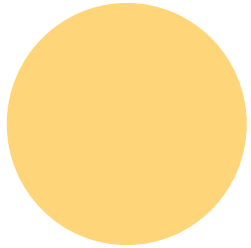
Data collected but not relevant to the question

No improvement even with more data

Predictive performance



Irrelevant information adds no value

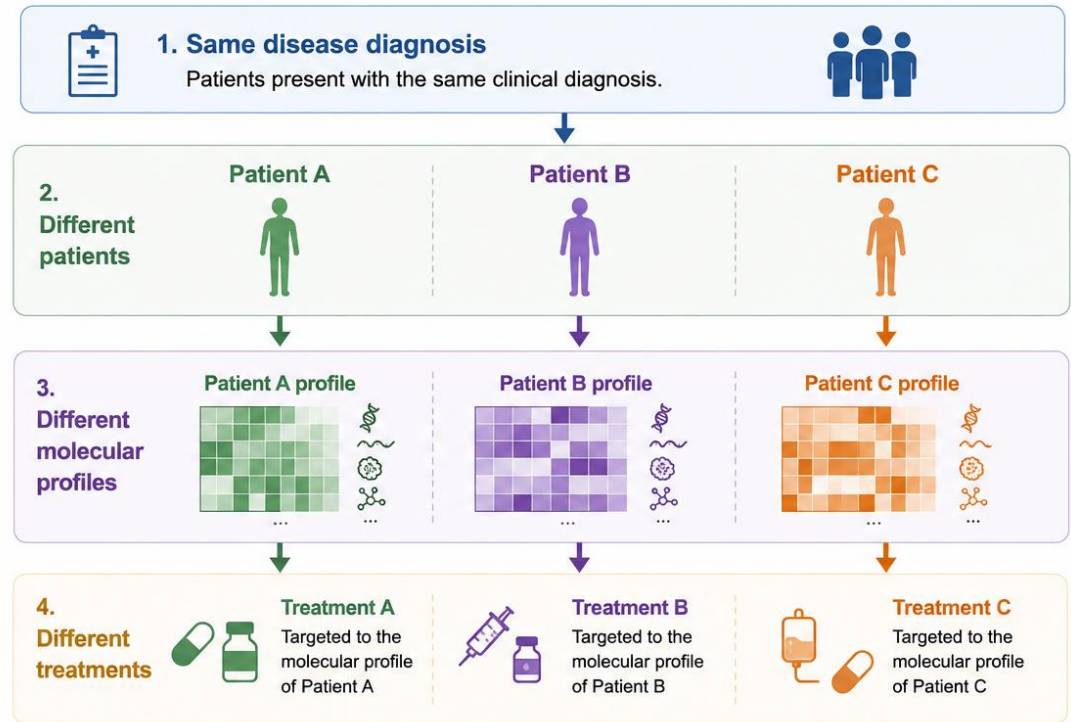


Personalised medicine



The personalised medicine dilemma

- Multi-omics is still largely a research tool.
- Few examples of clinical use.
- Technology is generally too expensive and lacks robustness.



Key takeaways

1



No single omics captures the whole biological system.

Each omic observes a different layer and provides a partial view.

2



Different omics operate on different scales and timescales.

They vary in causal distance, timing, physiological scale and measurement unit.

3



Integration requires biological, not just statistical, thinking.

Biology guides what to measure, how to integrate and how to interpret.

4



Experimental design is often more important than the integration method.

Thoughtful design maximises complementary information and minimises confounding.

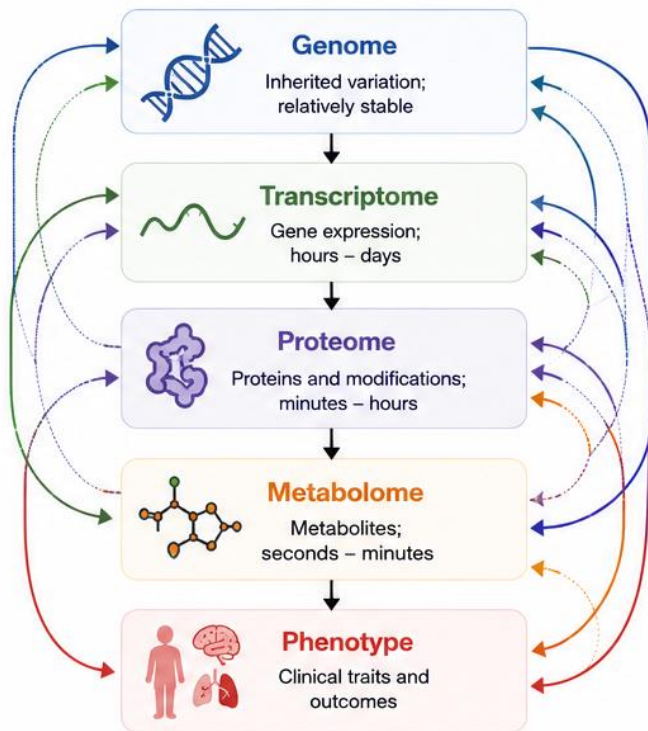
5



Mechanistic understanding remains the ultimate goal.

We aim to move from correlation to mechanism, and improve human health.

From causes to outcomes
Different lenses on the same system



Key dimensions to consider



Time scale

Biological processes occur from seconds to years.



Physiological scale

From molecules to cells, tissues, organs and individuals.



Information content

Overlap and orthogonality vary across modalities.



Causal distance

Predictive value depends on how close a layer is to the phenotype.

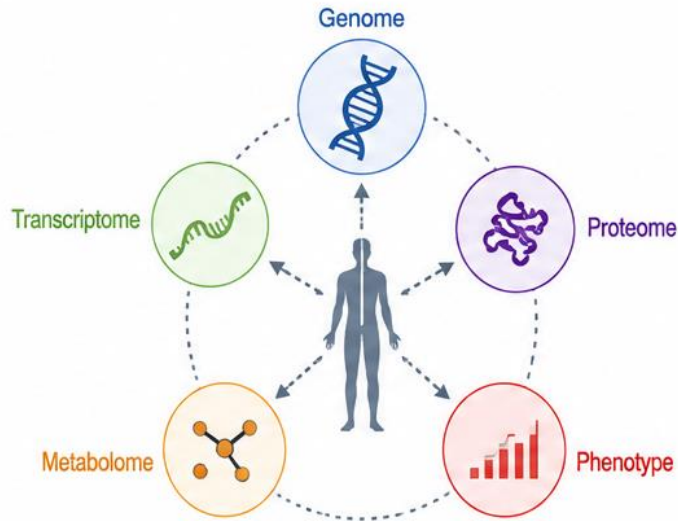


Measurement unit

Different units capture different aspects of biology.

Multi-omics can be about measuring things ...

... ideally it should be about measuring the right things:



Different layers, connected by biology,
Integration reveals the bigger picture.



1. At the right time scale

Capture biology at the time it actually changes.



2. At the right physiological scale

Measure at the scale where biology is organised and acts.



3. With the right mechanistic relationship

Choose measurements that are biologically connected.



4. With sufficient complementary information

Combine information that adds new insight, not just more of the same.



5. To answer a specific biological question

Let the question drive what, how and when you measure.